

Grace Hopper

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Summary

Computer scientist with 40+ years pioneering programming languages and compilers for the first digital computers. Strong at zero-to-one systems work, designing human-readable programming tools, and turning research into production. Deep experience with compiler design, machine instruction sets, and data processing standards, with a track record of driving adoption across industry and government and leading teams of engineers and officers.

Experience

U.S. Navy

Director, Navy Programming Languages Group

Aug 1967 - Aug 1986

- Recalled to active duty to standardize the Navy's high-level programming languages and compilers across the fleet.
- Drove definition and validation of COBOL standards and compiler conformance, unifying business computing across the Department of Defense.
- Developed compiler validation suites that certified consistent language behavior across dozens of vendor systems.
- Directed the Navy Programming Languages Group, coaching officers and engineers on software standards and long-term maintainability.
- Championed distributed networks of small computers over centralized mainframes, anticipating modern systems architecture.
- Delivered hundreds of lectures on computing efficiency and innovation to military, industry, and academic audiences.
- Retired as Rear Admiral in 1986, then the oldest serving officer in the U.S. Navy, and received the Defense Distinguished Service Medal.

Remington Rand

Director of Automatic Programming

Jun 1949 - Jul 1967

- Joined the Eckert–Mauchly Computer Corporation (later Remington Rand) as senior mathematician on the UNIVAC I, the first commercial electronic computer.
- Built the A-0 System, the first compiler, proving that a computer could translate human-readable symbolic code into machine instructions.
- Led development of successive automatic programming tools:
 - Designed FLOW-MATIC, the first English-like programming language, targeting business data processing.
 - Grew reusable subroutine libraries and compiler tooling that cut programming time from weeks to hours.
- Evangelized automatic programming to skeptical industry and government audiences, driving adoption of higher-level languages over hand-written machine code.

Harvard University

Research Fellow, Computation Project

Jul 1944 - Jun 1949

- Served as one of the first three programmers of the Harvard Mark I, computing critical ballistics and mathematical tables for the U.S. Navy during World War II.
- Authored the 561-page *Manual of Operation for the Automatic Sequence Controlled Calculator*, one of the earliest comprehensive references on programming a digital computer.
- Programmed the Mark II and Mark III and logged the original computer “bug,” coining terminology still used across software engineering today.

Vassar College

Associate Professor of Mathematics

Sep 1931 - Jun 1943

- Taught undergraduate mathematics and built early coursework connecting abstract theory to practical computation and probability.

Skills

Compiler Design A-0 System FLOW-MATIC COBOL Machine Code Assembly Symbolic Notation Subroutine Libraries
Harvard Mark I Harvard Mark II UNIVAC I Punched Cards Magnetic Tape Data Processing Standardization (ANSI / DoD)
Numerical Analysis Linear Algebra Calculus Differential Equations Probability Debugging Technical Writing

Selected Projects

[A-0 System](#)

The first compiler for an electronic computer, translating symbolic mathematical code into executable machine instructions. Established the foundation for automatic programming and reusable subroutine libraries.

UNIVAC I Machine Code Symbolic Notation Subroutine Libraries

[FLOW-MATIC](#)

The first programming language to use English-like statements. Served as the primary inspiration for COBOL.

UNIVAC I English-like Syntax Data Processing Compiler Design

[COBOL](#)

Common Business-Oriented Language, a portable and English-readable standard for business computing. Adopted across industry and government, it still runs critical systems worldwide.

English Syntax Data Processing Cross-Platform Standardization

[The “Nanoseconds” Lecture](#)

Signature talk explaining computing latency by handing out 11.8-inch wires representing the distance light travels in a nanosecond. A widely cited primer on efficiency for engineers and admirals alike.

[The First Computer “Bug”](#)

The original computer “bug” — a moth trapped in a Harvard Mark II relay — [logged and taped into the operations notebook](#), popularizing the terms “bug” and “debugging” still used across software engineering today.

Education

Vassar College, Bachelor of Arts, Mathematics and Physics

Sep 1924 - Jun 1928

Yale University, Doctor of Philosophy, Mathematics

Sep 1930 - Jun 1934